Does the training top-k matter? Sweeping the QFT compression rate

Question. Every trained QFT cell in this repository optimises `MSELoss(k)` — mean-squared reconstruction error after keeping the top-k transform coefficients by magnitude — with k fixed at 10% of the 2^{m+n} coefficients ($k = \mathrm{round}(2^{m+n} \cdot 0.1)$). Evaluation, by contrast, reports PSNR at four keep-ratios $\rho \in \{0.05, 0.10, 0.15, 0.20\}$. The training objective therefore optimises one compression rate while we deploy at several. Two questions follow: (i) is the fixed 10% training top-k actually optimal, and (ii) is each eval rate ρ best served by a QFT **trained at a matching** top-k ($k \approx \rho$)?

Method. We train the analytic-init `QFTBasis(m, n)` at four training top-k ratios $r \in \{0.05, 0.10, 0.15, 0.20\}$ and evaluate every trained operator at all four eval keep-ratios — the full 4×4 train-k $\times$ eval-ρ matrix. Each run is identical to the headline `qft` cell except for the training top-k: analytic QFT init, the `generalized` preset, 1008 steps (`--epochs 112`, no early stopping), seed 42. Two datasets: DIV2K-8q ($m = n = 8$, 256×256) and QuickDraw-5q ($m = n = 5$, 32×32).

Sanity check. The $r = 0.10$ DIV2K run reproduces the headline `qft` cell **exactly** — 25.093, 27.572, 29.532, 31.294 dB at $\rho = 0.05, 0.10, 0.15, 0.20$, $\Delta = 0.000$ dB at every ratio — confirming the sweep faithfully replicates the headline training pipeline and that the headline number was trained at 10%.

DIV2K-8q ($m = n = 8$)

train top-k	$\rho = 0.05$	$\rho = 0.10$	$\rho = 0.15$	$\rho = 0.20$
$r = 0.05$	24.912	27.304	29.201	30.906
$r = 0.10$ (default)	25.093	27.572	29.532	31.294
$r = 0.15$	24.992	27.456	29.405	31.151
$r = 0.20$	**25.235**	**27.811**	**29.841**	**31.659**

On DIV2K the **highest** training top-k wins at every eval rate: $r = 0.20$ beats the default $r = 0.10$ by +0.14, +0.24, +0.31, +0.37 dB. The matching rule (ii) is wrong — a single training top-k (0.20) dominates the whole column set, including the low-rate columns. Notably, $r = 0.20$ from the analytic init reaches 31.659 dB at $\rho = 0.20$, essentially the same flat-valley floor (≈ 31.66 dB) that identity-init QFT reaches at $r = 0.10$ — two routes to the same endpoint.

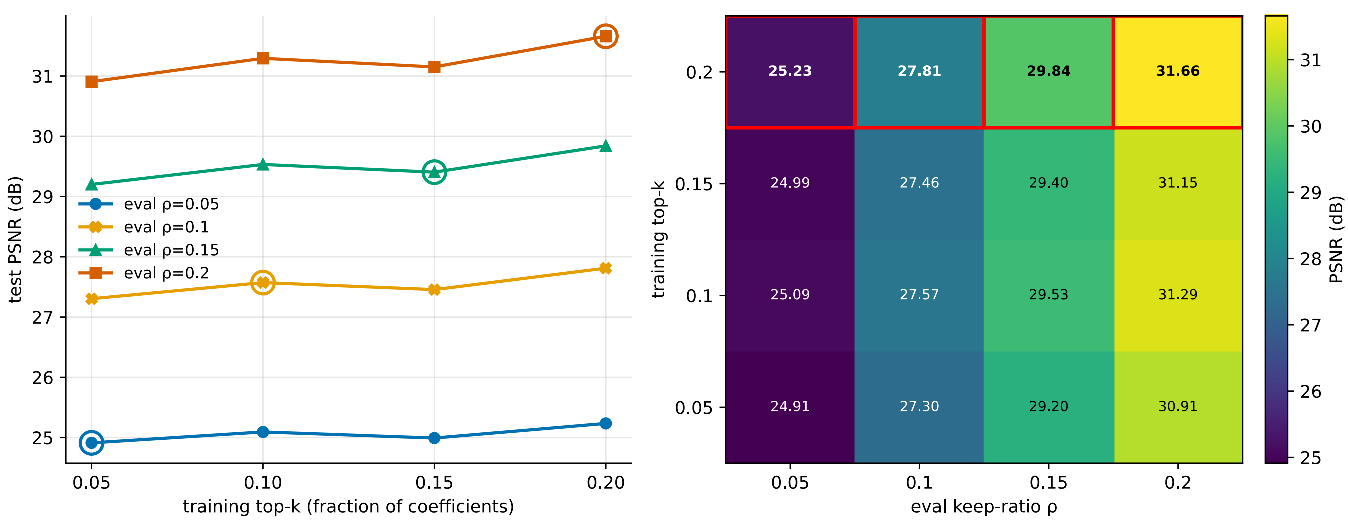

Figure 1: DIV2K-8q. Left: test PSNR vs training top-k, one line per eval keep-ratio; the ringed point on each line is the **matching** run ($r = \rho$). The rings are mostly **not** the peaks — each line keeps climbing past its matching point up to $r = 0.20$. Right: the full train-k $\times$ eval-ρ matrix; the best training top-k per eval column is boxed in red (the entire $r = 0.20$ row). The small dip at $r = 0.15$ is within single-seed basin noise (see **Caveats**).

QuickDraw-5q ($m = n = 5$)

train top-k	$\rho = 0.05$	$\rho = 0.10$	$\rho = 0.15$	$\rho = 0.20$
$r = 0.05$	**16.198**	**18.927**	**21.268**	23.443

train top-k	$\rho = 0.05$	$\rho = 0.10$	$\rho = 0.15$	$\rho = 0.20$
$r = 0.10$ (default)	16.196	18.921	21.267	**23.447**
$r = 0.15$	15.801	18.372	20.560	22.567
$r = 0.20$	15.800	18.370	20.557	22.568

QuickDraw shows the **opposite** trend: the **lowest** training top-k wins. $r = 0.05$ and $r = 0.10$ are tied and best at every rate; $r = 0.15$ and $r = 0.20$ are uniformly ≈ 0.7–0.9 dB worse (e.g. 23.447 vs 22.568 dB at $\rho = 0.20$). Again the matching rule fails — a single low training top-k dominates, rather than k tracking ρ.

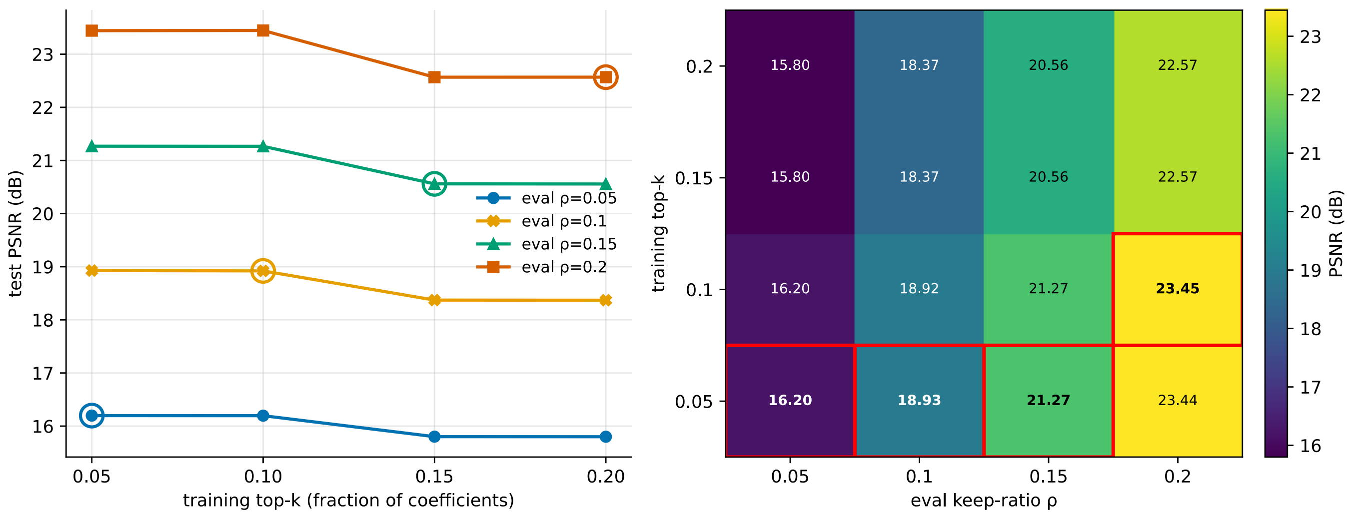

Figure 2: QuickDraw-5q. Same panels as above. Here the lines **fall** with training top-k: the best operator is trained at $r \in \{0.05, 0.10\}$, and the $r = 0.15/0.20$ runs sit a clear step lower at every eval rate.

What this shows

- **The fixed 10% training top-k is not optimal.** It is too low on DIV2K (where $r = 0.20$ gains up to $+0.37$ dB) and at the high end of the good range on QuickDraw (where $r \in \{0.05, 0.10\}$ tie and $r \geq 0.15$ loses ≈ 0.8 dB). The training top-k is a real hyperparameter, not a free choice.
- **"Train at the rate you deploy at" is the wrong rule.** On neither dataset does matching $k \approx \rho$ maximise PSNR at ρ. A single, dataset-specific training top-k is best across **all** eval rates simultaneously — high (0.20) for DIV2K, low (≤ 0.10) for QuickDraw.
- **The optimum is dataset-dependent and opposite across scales.** The $m = 8$ natural-image dataset prefers a broader training objective; the $m = 5$ sparse-sketch dataset prefers a narrower one. A plausible reading: QuickDraw bitmaps concentrate energy in very few coefficients, so a wide top-k spends gradient budget fitting near-zero coefficients; DIV2K photographs have heavier coefficient tails, so a wider top-k supervises more of the signal that actually matters at $\rho = 0.20$.

Caveats

Single seed. Every cell is one run at seed 42 on the headline 1008-step budget. The DIV2K $r = 0.15$ dip below $r = 0.10$ is the clearest symptom: the cosine LR schedule is tied to the total epoch count, and past step ≈ 700 the top-k MSE valley is very flat, so different runs can settle in slightly different basins (a caveat inherited from the headline protocol). The **direction** of the effect is consistent within each dataset across all four eval rates, but the sub-0.1 dB orderings (e.g. $r = 0.05$ vs $r = 0.10$ on QuickDraw) should not be over-read. A multi-seed repeat is the natural follow-up.

Reproducibility.

```
python experiments/qft_topk_sweep.py --gpu 0 --dataset div2k_8q
python experiments/qft_topk_sweep.py --gpu 0 --dataset quickdraw_5q
python tools/render_topk_sweep.py --dataset div2k_8q
python tools/render_topk_sweep.py --dataset quickdraw_5q
```

Training top-k count: `experiment_utils.train_k_for(m, n, r)` $= \max(1, \mathrm{round}(2^{m+n} \cdot r))$. Cells land at `results/training/qft_topk_sweep/<dataset>/_runs/train_k<pct>/`; the full matrix is in each dataset's `manifest.json`.